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Multi-Modal Sensor Fusion for Smart Home HAR

23CSE399 – Project Phase 1 | Panel Review 1

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1. Originality of Use Case (1/2)

Rubric: Novel Use Case & Applicability

Context

Smart home technology has huge potential for elderly care and remote health monitoring. The key is Human Activity Recognition (HAR) — enabling the house to understand what a person is doing (cooking, exercising, or falling) to provide help. Current systems are stuck: either accurate but privacy-invasive, or private but not smart enough.

Limitation of Single Sensors

- **Cameras:** Accurate but violate privacy in personal spaces.
- **Wearables:** Good for motion tracking but annoying to wear 24/7. If forgotten, the system fails.
- **WiFi/RF Sensors:** Device-free but get confused easily by environmental noise.
- **Ambient Sensors:** Private but cannot distinguish between complex actions.

Project Objective

This project proposes a Multi-Modal Sensor Fusion Framework combining Ambient, Inertial, and RF sensors to achieve five goals current systems fail at:

- **Privacy:** High accuracy without Cameras.
- **Reduced Wearables:** Rely primarily on environmental sensors.
- **Multitasking:** Recognizing Parallel Activities.
- **Multi-User:** Handling concurrent individuals.
- **Identification:** Unique Person Identification.

1. Originality of Use Case (2/2)

Rubric: Novel Use Case & Applicability

1. Robust Data Capture & Sensor Fusion

Design a system to capture and fuse data from diverse, non-intrusive sensors (RF, ambient, minimal wearables) while minimizing data loss. Address recognizing **Concurrent activities**, **alternating activities**, **multi-user environments**, and **accurate person identification**.

2. Multi-Modal Architecture & Model Building

Develop and evaluate deep learning models for multi-modal data. Compare **independent models per sensor modality** vs. an **aggregated model** to find the optimal architecture for varying data frequencies.

3. Real-Time Actuation & Application

Apply the recognized multi-modal data to an assisted-living / elderly care / health care use case. Provide automated, **real-time actuation** — e.g., **triggering an alert** or specific home response when a fall or anomaly is detected.

2. Benchmarking – Dataset Survey (1/2)

Rubric: Clear Comparison – Ambient & Wearable Datasets

Type	Dataset	Description & Sensors	Critical Limitation
Ambient	CASAS (Cook et al.)	Gold standard smart home: PIR motion, door, and temperature sensors in real apartments.	Identity ambiguity – cannot tell *who* triggered a sensor in multi-resident homes.
Ambient	ARAS (Alemdar et al.)	20 ambient sensors in 2 real houses with multiple residents.	High false positives from pets/visitors; no ground truth for overlapping activities.
Wearable	UCI-HAR (Anguita et al.)	Smartphone accel & gyroscope (waist-worn) for 6 basic ADLs.	Location blind – detects "Walking" but not *where* (kitchen vs. bedroom).
Wearable	Opportunity (Roggen et al.)	72 sensors on body & objects capturing gestures (opening doors, drinking).	Impractical – users must wear a full body sensor suit in daily life.
Wearable	UniMiB-SHAR (Micucci et al.)	Smartphone accelerometer focused on fall detection and basic activities.	Limited granularity – good for falls, poor at fine-grained ADLs (e.g., eating).

2. Benchmarking – Dataset Survey (2/2)

Rubric: Clear Comparison – WiFi / RF Datasets

Type	Dataset	Description & Sensors	Critical Limitation
WiFi	Widar 3.0 (Zheng et al.)	Large-scale WiFi CSI dataset for gesture recognition using commodity routers.	Single-user focus – accuracy drops when multiple users move simultaneously.
WiFi	CSI-Bench (Wang et al.)	Robustness benchmark for WiFi sensing across different rooms and environments.	Domain shift – models trained in Room A fail in Room B without retraining.
WiFi	WIMANS (Gu et al.)	Multi-user dataset: dual-band WiFi CSI with synchronized camera ground truth.	Heavy computation needed to separate mixed signals from multiple users.
WiFi	SignFi (Ma et al.)	Fine-grained finger gesture / sign language recognition using WiFi CSI.	Short range ($<$1m) only; not suitable for whole-home activity tracking.
WiFi	TED-Net (Wang et al.)	Privacy-preserving fall detection using WiFi channel state information.	Hardware-specific – relies on router configurations hard to deploy universally.

2. Benchmarking – Competitor Analysis

Rubric: Clear Comparison – Commercial Solutions

Competitor	Primary Focus	Research Gap
Smart Home Hubs (Google Nest, Alexa)	Voice command & rule-based automation (motion → lights on).	Context-unaware: cannot distinguish "Fall" from "Lying Down".
Wearable Tech (Fitbit, Apple Watch)	Biometrics & fitness: heart rate, steps, outdoor GPS.	Indoor blind: no awareness of room or home-environment context.
Security Systems (ADT, Ring)	Perimeter safety via door sensors and cameras.	Privacy-invasive cameras; non-camera sensors are binary only.
Elderly Monitors (LifeAlert)	Emergency response via manual button press after a fall.	Purely reactive: triggers *after* accident; cannot predict incidents.
WiFi Sensing (Origin Wireless)	Device-free presence and breathing detection via WiFi.	Identity confusion: signals mix; cannot track multiple people reliably.

3. Technical Architecture – System Overview

Rubric: End-to-End Data Flow & Tools

Sensor Inputs (Multi-Modal):

- **Ambient:** PIR motion, door & temperature sensors.
- **RF / WiFi CSI:** Device-free activity & presence detection via Bazel Kit.
- **Minimal Wearable:** IMU (accel + gyro) for person identity tagging.

End-to-End Data Flow:

Sensors → **Preprocessing** → **Feature Extraction**

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Fusion Model → **Activity Classification**

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Real-Time Actuation

Core Model Components:

- **Dual-Branch CNN-LSTM:** CNNs extract spatial features; LSTMs capture temporal dependencies across heterogeneous sensor streams.
- **Attention-Enhanced Fusion:** Dynamically weights modality importance (e.g., wearable prioritised for fine motor tasks) before final classification.
- **Preprocessing Pipeline:** Temporal alignment + sliding-window segmentation to standardise varying sampling rates.

Output / Actuation:

- Real-time smart home responses (e.g., fall alert, room automation).
- Edge inference deployed on Bazel Kit hardware.

3. Technical Architecture – System Diagram

Rubric: End-to-End Data Flow & Tools

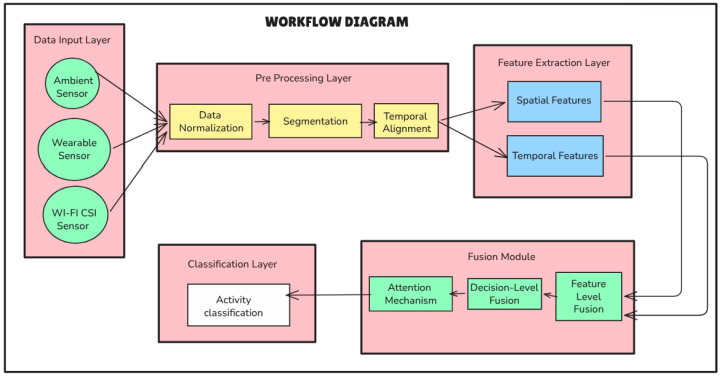


Fig. Proposed Multi-Modal Sensor Fusion Architecture

Key Design Decisions:

- **No camera used** – RF + Ambient sensors preserve user privacy completely.
- **Attention mechanism** ensures each modality contributes proportionally based on the activity type being recognised.

3. Technical Architecture – Phase 1: Completed Work (1/2)

Rubric: End-to-End Data Flow & Tools

Dataset Preparation (CASAS Datasets: Milan, Cairo, Kyoto2, Kyoto3, Kyoto4):

- **Step 1: Time Ordering** – Sensor events are sorted chronologically to understand the sequence of actions.
- **Step 2: Activity Grouping** – Events are grouped by activity start and end times, converting random logs into sequence classification tasks.
- **Step 3: Numerical Format** – Sensors are assigned indices, creating binary vectors (1 for active, 0 for inactive) at each time step.
- **Step 4: Sequence Padding** – Sequences are standardized to a fixed length by padding shorter activities with zeros and trimming longer ones.
- **Step 5: Tensor Generation** – The final sensor sequences (X) and activity labels (y) are saved as `.npy` files for efficient LSTM training.

3. Technical Architecture – Raw Data Structure

Rubric: End-to-End Data Flow & Tools

The CASAS dataset consists of two primary data streams that must be synchronized during preprocessing:

Stream 1: Continuous Sensor Events

Time	Sensor	State
12:18:44	M16	ON
12:18:46	M17	OFF
12:28:50	D12	OPEN

Stream 2: Activity Ground Truth

Start Time	End Time	Activity
12:00:00	12:08:00	Cook
22:00:00	06:00:00	Sleep

Note: The preprocessing pipeline maps Stream 1 into the time windows defined by Stream 2 to create training tensors.

3. Technical Architecture – Model Performance Summary

Rubric: End-to-End Data Flow & Tools

Model	Performance Trend	Complexity
Uni-LSTM	Strong baseline	Low
Bi-LSTM	Slight improvement	Moderate
Casc-LSTM	Inconsistent improvement	Higher
Ens2-LSTM	Best on large datasets	Moderate
CascEns-LSTM	Marginal gains	High

3. Technical Architecture – Phase 1: Completed Work (2/2)

Rubric: End-to-End Data Flow & Tools

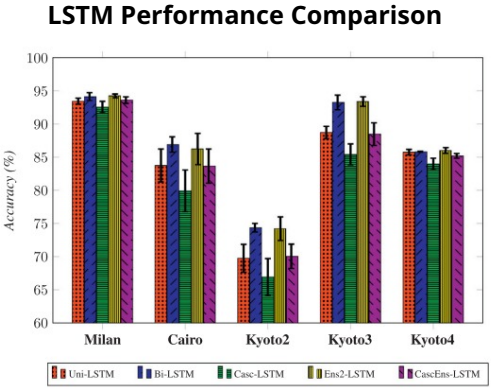


Fig. 9. Averaged accuracy and standard deviation of each dataset over all three folds.

Fig. Averaged accuracy and standard deviation of each dataset over all three folds.

3. Technical Architecture – Phase 2: Planned Work

Rubric: End-to-End Data Flow & Tools

Multi-Modal Data Collection:

- Collect and synchronise **WiFi CSI data** alongside ambient sensor logs.
- Capture **minimal wearable inertial data** (accel + gyro) via Bazel Kit.
- Construct a custom multi-modal dataset for training and evaluation.

Hybrid Architecture Development:

- Extend validated LSTM backbone into **dual-branch CNN-LSTM**.
- Integrate **attention-based fusion** for adaptive modality weighting.
- Compare independent modality models vs. aggregated fusion model.

Hardware Deployment & Actuation:

- Deploy trained fusion model on **Bazel Kit** edge hardware.
- Enable **real-time inference** for assisted-living scenarios.
- Trigger automated smart home responses (fall alert, room-based automation).

Evaluation Targets:

- Accuracy and F1-score per activity class.
- Inference latency on edge hardware.
- Cross-environment and cross-user generalisation.

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Thank You

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